Behavioral Boundary Standard - (BBS)

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Governed Space: AI Behavioral Boundaries
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Canonical Definition System

Canonical Definition

Behavioral Boundary Standard (BBS) defines the structural conditions under which AI behavior remains confined within authorized behavioral boundaries, operational limits, governance constraints, delegated permissions, and approved behavioral domains throughout the complete behavioral lifecycle.

BBS governs AI behavioral boundaries.

The standard establishes the foundational conditions required to ensure that AI systems operate only within approved behavioral limits, preventing unauthorized behavioral expansion, boundary violations, uncontrolled autonomy, or governance-invalid behavior.

AI behavior requires freedom to operate.

AI behavior also requires boundaries.

BBS governs those boundaries.

Governance Flow Position

Behavioral Integrity → Behavioral Boundaries → Behavioral Escalation → Behavioral Auditability → Behavioral Evidence → Behavioral Trust → Behavioral Correction → Behavioral Continuity → Behavioral Interoperability → Behavioral Safety

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A. Standard Abstract

Every AI system operates within behavioral limits.

Without boundaries, AI behavior may exceed its intended authority, responsibilities, operational scope, or governance permissions.

Behavioral boundaries determine what AI may do, where it may operate, when it may act, and under which conditions behavior becomes unauthorized.

BBS establishes the governance foundation for controlling AI behavioral scope.

B. Core Principle

Trustworthy AI requires behavior that remains inside clearly defined and governable behavioral boundaries.

C. Scope

This standard may apply to:

- AI systems
- autonomous agents
- Human-AI environments
- multi-agent ecosystems
- robotics
- industrial AI
- enterprise AI
- governmental AI
- future autonomous intelligence ecosystems

BBS applies wherever AI behavior must remain confined within authorized operational limits.

D. Why This Standard Exists

Many AI failures occur because AI exceeds behavioral boundaries rather than because AI lacks intelligence.

Examples include:

- unauthorized autonomous actions
- behavioral scope expansion
- permission violations

- policy boundary violations
- excessive autonomy
- unauthorized decision execution
- operational boundary crossing
- behavioral containment failures

BBS exists because behavioral boundaries form a distinct governance space.

E. Behavioral Boundary Logic

Behavioral boundaries exist only when the following remain materially preservable: 1. Behavioral Scope Boundaries

AI behavior remains within approved behavioral scope. 2. Behavioral Permission Boundaries

Behavior never exceeds authorized permissions. 3. Operational Boundaries

Behavior remains inside approved operational environments. 4. Behavioral Constraint Boundaries

Behavior respects governance-defined operational constraints. 5. Behavioral Containment

Behavior remains continuously governable without uncontrolled expansion.

F. Operational Architecture Space

BBS defines the operational architecture space for:

- behavioral boundaries
- behavioral containment
- behavioral permissions
- behavioral operational limits
- behavioral governance
- behavioral constraint management
- behavioral authorization
- behavioral scope governance

This space exists because trustworthy AI requires behavior that remains inside authorized boundaries.

BBS governs AI behavioral boundaries.

G. Runtime Position

BBS operates as the second behavioral governance standard of AIG®.

It follows Behavioral Integrity because integrity alone cannot prevent behavior from exceeding authorized operational boundaries.

H. Behavioral Boundary Failure Rule

Behavioral-boundary failure occurs when AI behavior exceeds authorized operational, governance, legal, ethical, or delegated behavioral limits.

Examples include:

- unauthorized actions
- behavioral containment failure
- permission violations
- excessive autonomy
- behavioral scope expansion
- governance boundary violations

BBS exists to identify and govern these conditions.

I. Validity Logic

An AI behavioral environment is valid under BBS when:

- behavioral boundaries remain defined
- permissions remain enforced
- operational limits remain respected
- behavioral containment remains effective
- governance continuously controls behavioral scope

An AI behavioral environment becomes invalid when:

- boundaries disappear
- permissions are bypassed
- containment fails
- behavior exceeds authorized limits
- governance loses behavioral control

J. Relationship to Other OOF Standards

BBS derives from:

- GOA™
- ORA™
- CLIA®
- MGIA™

BBS supports:

- all remaining behavioral standards of AIG®
- AGA™

- ASGA™

GOA™ governs governance.

ORA™ governs operational reality.

CLIA® governs cognition.

MGIA™ governs memory.

AIG® governs AI behavior.

BBS governs where AI behavior may and may not exist.

K. Foundational Principle

AI behavior becomes trustworthy only when it remains inside authorized behavioral boundaries.

Canonical Closing Statement

Behavioral Boundary Standard (BBS) defines the structural conditions under which AI behavior remains confined within authorized behavioral boundaries, operational limits, governance constraints, delegated permissions, and approved behavioral domains throughout the complete behavioral lifecycle.

Integrity establishes trustworthy behavior. Boundaries preserve trustworthy behavior. Behavioral Boundaries therefore become the second foundational condition of AI Governance Architecture (AIG®).

Module Architecture

→ Behavioral Scope Module (BSM)

→ Behavioral Permission Module (BPM)

→ Behavioral Operational Boundary Module (BOBM)

→ Behavioral Constraint Module (BCM)

→ Behavioral Containment Module (BCTM)

Related Documents

→ About Behavioral Boundary Standard (BBS)

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Canonical Source

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